Einstein's Missed Connection: Block Universe Implies Superdeterminism

How Three of Einstein's Own Commitments Entail the Interpretation He Never Considered

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Abstract

Einstein held three commitments throughout his career: the block universe (all of spacetime exists as a four-dimensional manifold, a direct consequence of special and general relativity), determinism (physical processes are governed by causal law, not chance), and hidden variables (quantum systems possess definite properties prior to measurement, as argued in EPR). These three commitments jointly entail superdeterminism: the denial of statistical independence between measurement settings and hidden variables. In a block universe, measurement choices are fixed features of the spacetime manifold, not free inputs. They share causal ancestry with every other event, including the hidden variables they are supposed to be independent of. Bell's theorem (1964) formalized statistical independence as the precise assumption whose denial permits local deterministic accounts of quantum correlations. Einstein died in 1955, nine years before Bell identified the exact escape hatch that Einstein's own ontology provides. Bell's theorem is often read as ruling out local determinism. It does not. It proves that statistical independence must fail if local determinism is to hold. Einstein's block ontology perfectly fits both many worlds and superdeterminism. For many worlds, Einstein was trying to complete quantum mechanics with additional hidden variables, with the wave equation as emergent. With hypergraphs, he may still be right yet incomplete about that. What's the difference? Causality. Graphs would be more fundamental. This paper traces the historical path by which Einstein's three commitments were separated rather than unified, and argues that the physics community's dismissal of superdeterminism as conspiratorial rests on importing libertarian free will into a block universe that excludes it by construction.

Keywords: Einstein, superdeterminism, block universe, statistical independence, Bell's theorem, EPR, determinism, history of physics

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1. Introduction

The standard narrative of quantum foundations runs as follows. Einstein, Podolsky, and Rosen (1935) argued that quantum mechanics is incomplete: entangled particles must carry definite values prior to measurement. Bohr replied that complementarity dissolves the paradox. Bell (1964) proved that no local hidden-variable theory satisfying statistical independence can reproduce quantum predictions. Experiments confirmed Bell inequality violations (Aspect et al., 1982; Hensen et al., 2015). Conclusion: local realism fails.

This narrative omits a logical possibility that Einstein's own commitments make available. Einstein was not merely a local realist. He was a block universe determinist. Special relativity eliminates absolute simultaneity. General relativity completes the picture: spacetime is a four-dimensional manifold in which all events, past, present, and future, coexist. Einstein endorsed this view explicitly. In a 1955 letter to Michele Besso's family, written weeks before his own death, he wrote that the distinction between past, present, and future is a stubbornly persistent illusion.

In this block, measurement choices are not free variables. They are events within the manifold, determined by the same initial conditions that determine everything else. Statistical independence, the assumption that measurement settings are uncorrelated with hidden variables, is not a physical law. It is an assumption about the relationship between two subsets of events in the block. In a fully deterministic block universe, this assumption is false by construction.

Einstein never made this connection. This paper asks why, and traces the consequences of the omission.

2. Einstein's Three Commitments

2.1 Block Universe

Special relativity (1905) eliminates absolute simultaneity: events that are simultaneous in one reference frame are not simultaneous in another. This makes the present observer-dependent. If "now" is relative, there is no objective division of spacetime into past and future. All events exist equally. General relativity (1915) deepens this: spacetime is a dynamical manifold, and the field equations describe its geometry without privileging any temporal slice. The resulting ontology is the block universe: a four-dimensional structure in which time is a dimension, not a process.

Einstein accepted this. He did not treat the block universe as a mathematical convenience or a coordinate artifact. He treated it as the nature of reality.

2.2 Determinism

Einstein's opposition to quantum indeterminacy is well documented. "God does not play dice" is the popular summary, though his actual position was more precise: he held that a complete physical theory must specify the evolution of a system without irreducible randomness. The EPR paper frames this as an argument for hidden variables: if a measurement outcome can be predicted with certainty without disturbing the system, then the corresponding physical quantity has a definite value prior to measurement.

Determinism in a block universe is not merely a preference. It is a structural consequence. If all events coexist in a four-dimensional manifold, every event is fixed. There is no objective process by which an indeterminate event becomes determinate. “Becoming” requires a privileged present, and the block universe has none.

2.3 Hidden Variables (EPR)

The EPR argument (1935) is standardly read as an argument for local realism. But it is better understood as an argument for completeness: quantum mechanics, in its Copenhagen form, fails to describe physical reality completely. The correlations between entangled particles are too strong to be explained by shared classical information, yet EPR assumes they must have a local explanation. The conclusion: QM is incomplete, and hidden variables must exist.

2.4 The Entailment

Block universe: all events are fixed. Determinism: no irreducible randomness. Hidden variables: properties exist prior to measurement. In this ontology, measurement settings are not free inputs chosen independently of the system. They are fixed events in the manifold, determined by the same initial conditions that determine the hidden variables. Statistical independence, Bell’s assumption that the probability distribution of hidden variables is independent of measurement settings, is false. Not because of conspiracy, but because of geometry. In a block universe, nothing is independent of anything else. Everything is part of one structure. Chapter 1 and chapter 20 of a novel are correlated not because chapter 20 sends a signal back, but because they are part of one book.

3. Why Einstein Missed It

Three factors explain the omission.

3.1 Locality as sacred. Einstein treated locality, the principle that physical influences propagate at or below the speed of light, as non-negotiable. His objection to quantum mechanics was precisely that entanglement seemed to violate it. He sought a theory that preserved locality by completing quantum mechanics with hidden variables. Superdeterminism preserves locality, but Einstein never framed the question in terms of statistical independence. The concept had not been formalized.

3.2 Bell had not yet published. Bell’s theorem (1964) is the crucial step. It identifies statistical independence as a formal assumption and proves that its satisfaction, combined with locality, constrains correlations to the Bell inequality. Without Bell, there is no precise assumption to deny. Einstein could not have targeted statistical independence because the target did not yet exist in explicit form.

3.3 Superdeterminism appeared conspiratorial. Even without Bell's formalization, Einstein could in principle have noted that determinism correlates everything with everything. But the implication, that the initial conditions of the universe encode correlations between future measurement settings and particle properties, seems to require an implausible conspiracy. This objection persists in modern physics (Zeilinger, 1999; Maudlin, 2014). Under algorithmic probability (Solomonoff, 1964), the objection conflates specificity with complexity. Simple deterministic rules produce highly specific correlations at low descriptive cost. The patterns are intricate, but the generators are short. A fractal coastline is specific, but the rule is simple. The initial conditions of a superdeterministic universe can be informationally compact while producing arbitrary apparent complexity. Moreover, in a block universe, the phrase "initial conditions" is misleading. There is no temporal beginning at which conditions are set. The block is a timeless structure. The correlations are geometric features of that structure, not arrangements made at a starting point. The conspiracy objection presupposes a temporal setup that the block universe excludes.

4. The Community's Inheritance

After Einstein's death, the physics community inherited his locality but not his block universe. The mainstream interpretation of Bell's theorem became: local realism fails. The block universe, though a direct consequence of Einstein's own relativity, was treated as philosophy rather than physics. Statistical independence was treated as obviously true, importing an assumption of libertarian free will, the ability to choose measurement settings independently of the physical state of the universe, into a physical theory that excludes it by construction.

This is the core irony. The community rejected superdeterminism on grounds that Einstein's own ontology invalidates. They defended measurement independence by appealing to experimenter freedom, but the block universe has no experimenter freedom. Every choice is a fixed event in the manifold. Rejecting superdeterminism in a block universe requires either abandoning the block universe or smuggling in a form of free will that relativity rules out.

5. Entanglement in the Block Universe

Entanglement is standardly presented as a puzzle about nonlocal correlations. Two particles interact, then separate. Their combined state is described by one wave function, not two independent ones. Measure one particle's spin along some axis, and you instantly know the other's, regardless of the distance between them. The correlations exceed any bound derivable from local hidden variables satisfying statistical independence.

In the block universe, the puzzle dissolves. The particles were never independent systems that later become mysteriously correlated. They are features of a single four-dimensional structure. The measurement outcomes, the measurement choices, and the particle creation event all coexist in the manifold. The correlations are not produced by a signal traveling between the particles. They are structural features of the block, like the relationship between two points on a single geometric surface.

This does not make entanglement trivial. The specific structure of quantum correlations, their violation of Bell inequalities, their monogamy properties, and their information-theoretic constraints, all require explanation. What the block universe provides is the correct ontological framing: the correlations are geometric, not causal. They do not require action at a distance because there is no distance in the relevant sense. The events are related by their position in a structure, not by signals passing between them.

6. Implications

6.1 For the history of physics. The standard narrative, that EPR was refuted by Bell and confirmed by Aspect, is incomplete. EPR's conclusion that quantum mechanics is incomplete may be correct. Bell proved that the completion cannot be local if statistical independence holds. But Einstein's own block universe implies that statistical independence does not hold. The correct reading: Bell proved that local hidden-variable theories require superdeterminism, and Einstein's ontology provides it.

6.2 For current foundations research. The dismissal of superdeterminism rests on an assumption, experimenter freedom, that is inconsistent with the block universe that the same community accepts as a consequence of relativity. This inconsistency should be addressed rather than ignored.

6.3 For algorithmic probability. Under the Simplicity Dominance Principle, superdeterministic initial conditions generated by simple rules are measure-favored, not measure-suppressed. The fine-tuning objection conflates specificity with complexity: the digits of pi are extraordinarily specific but generated by a short program. The block universe provides the ontological ground, and algorithmic probability provides the measure-theoretic ground, for taking superdeterminism seriously. Kolmogorov measured the complexity of individual objects. Solomonoff used it to predict which patterns to expect. Neither made the ontological claim. SDP does: simpler structures dominate the measure over all that exists, and observers find themselves in simpler ones by mathematical necessity, not luck. SDP makes Occam's razor a theorem rather than a heuristic.

7. Conclusion

Einstein held three commitments: block universe, determinism, and hidden variables. These three entail superdeterminism. He never connected them. Bell formalized the key assumption nine years after Einstein's death. The physics community inherited Einstein's locality without his block universe, and dismissed superdeterminism on grounds, experimenter freedom, that Einstein's own relativity excludes. The missed connection is not merely a historical curiosity. It is a live error in the foundations of physics, and correcting it reopens a path that the community prematurely closed.

References

Aspect, A., Dalibard, J., & Roger, G. (1982). Experimental realization of Einstein-Podolsky-Rosen-Bohm Gedankenexperiment. *Physical Review Letters*, 49(25), 1804-1807.

Bell, J. S. (1964). On the Einstein Podolsky Rosen paradox. *Physics Physique Fizika*, 1(3), 195-200.

Ciepielewski, G. S., Okon, E., & Sudarsky, D. (2023). On superdeterministic rejections of settings independence. *British Journal for the Philosophy of Science*, 74(2), 381-398.

Einstein, A., Podolsky, B., & Rosen, N. (1935). Can quantum-mechanical description of physical reality be considered complete? *Physical Review*, 47(10), 777-780.

Hensen, B., et al. (2015). Loophole-free Bell inequality violation using electron spins separated by 1.3 kilometres. *Nature*, 526, 682-686.

Palmer, T. N. (2024). Superdeterminism without conspiracy. *Proceedings of the Royal Society A*, 480.

Zeilinger, A. (1999). A foundational principle for quantum mechanics. *Foundations of Physics*, 29(4), 631-643.